

### Project Initialization and Planning Phase

|                      |                                                               |
|----------------------|---------------------------------------------------------------|
| <b>Date</b>          | 02 July 2026                                                  |
| <b>Team ID</b>       | TEAM-HDI-2026                                                 |
| <b>Project Title</b> | A Comprehensive Measure of Well-Being (HDI Prediction System) |
| <b>Maximum Marks</b> | 5 Marks                                                       |

### Project Proposal (Proposed Solution) report

The proposal report aims to assess and predict a country's human development status using machine learning, boosting the efficiency of socio-economic research and policy planning. It tackles the inefficiencies of manual HDI calculations, promising faster insights, accurate tier classifications, and better decision-making tools. Key features include a machine learning-based HDI classification model and a real-time interactive web interface for prediction.

### Project Overview

| Field            | Details                                                                                                                                                                                                                                       |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Objective</b> | The primary objective is to develop a predictive machine learning model that accurately classifies countries into four HDI tiers (Very High, High, Medium, Low) based on life expectancy, education, and income indicators.                   |
| <b>Scope</b>     | The project comprehensively assesses global development data, incorporates machine learning for HDI tier prediction, and provides a web-based interface for policymakers and researchers to simulate and evaluate human development outcomes. |

## Problem Statement

| Field       | Details                                                                                                                                                                                                                         |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description | Addressing the lack of accessible, real-time predictive tools for evaluating multi-dimensional socio-economic data (health, education, income) which currently requires complex manual calculations and static report analysis. |
| Impact      | Solving these issues will result in improved efficiency for policymakers and researchers, enabling data-driven decision-making, better resource allocation, and a deeper understanding of global development gaps.              |

## Proposed Solution

| Field        | Details                                                                                                                                                                                                                                                                                                           |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Approach     | Employing supervised machine learning classification techniques to analyze historical UNDP/World Bank data and predict a country's HDI tier based on input indicators.                                                                                                                                            |
| Key Features | <ul style="list-style-type: none"><li>• Implementation of a machine learning-based HDI tier classification model.</li><li>• Real-time decision-making via an interactive Flask web application.</li><li>• Interactive data visualizations to compare user inputs against global development benchmarks.</li></ul> |

## Resource Requirements

| Resource Type           | Description                           | Specification/Allocation                                                                                              |
|-------------------------|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Hardware                |                                       |                                                                                                                       |
| Computing Resources     | CPU specifications                    | Intel Core i3 or above                                                                                                |
| Memory                  | RAM specifications                    | Minimum 4 GB (8 GB recommended)                                                                                       |
| Storage                 | Disk space for data, models, and logs | Minimum 10 GB free space                                                                                              |
| Software                |                                       |                                                                                                                       |
| Frameworks              | Python web frameworks                 | Flask                                                                                                                 |
| Libraries               | Additional libraries                  | scikit-learn, pandas, numpy, matplotlib, seaborn                                                                      |
| Development Environment | IDE                                   | Jupyter Notebook, VS Code / PyCharm, Anaconda Navigator                                                               |
| Data                    |                                       |                                                                                                                       |
| Data                    | Source, size, format                  | UNDP Human Development Reports / World Bank Open Data, CSV format, historical data for 190+ countries across multiple |